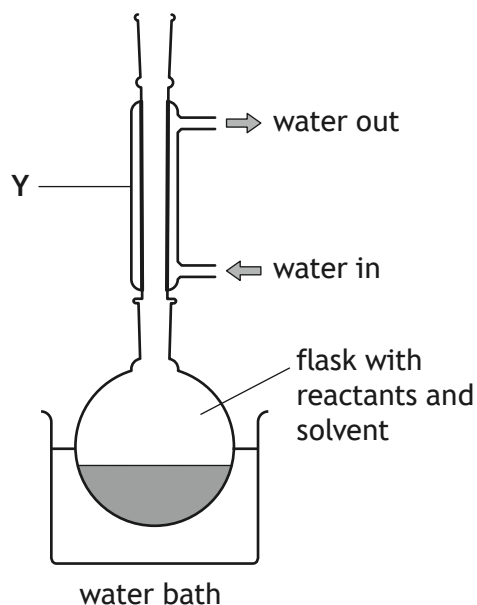


11. A Grignard reaction involves reacting compounds containing a carbonyl group with a Grignard reagent to make alcohols.

- (a) The first step in the reaction involves heating the reactants in a solvent to form a Grignard reagent.



- (i) Suggest why a water bath is used to heat the reaction.

1

- (ii) Name the piece of apparatus labelled Y in the diagram.

1

[Turn over

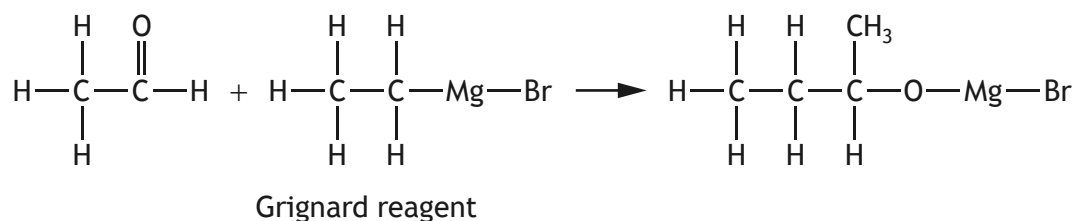


* X 8 1 3 7 6 0 1 3 1 *

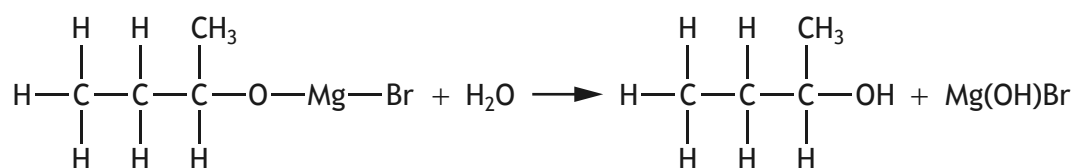
11. (continued)

(b) Ethanal reacts with a Grignard reagent in two steps.

step 1



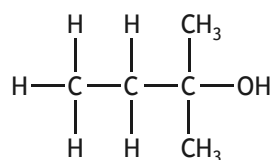
step 2



(i) Suggest a name for the reaction shown in step 1.

1

(ii) The alcohol that is made when propanone reacts with the same Grignard reagent is shown.



Name this alcohol.

1



11. (b) (continued)

- (iii) Draw a structural formula for the alcohol formed when the Grignard reagent $\text{CH}_3\text{CH}_2\text{MgBr}$ reacts with pentan-2-one.

1

[Turn over]



* X 8 1 3 7 6 0 1 3 3 *